

Multivariable Calculus

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Review: In Chapter 11, we discuss vector-valued functions of a single (scalar) variable, e.g. $\underline{r}(t) = x(t)\underline{\hat{i}} + y(t)\underline{\hat{j}} + z(t)\underline{\hat{k}}$.

(we input a scalar quantity, and obtain a curve in $\mathbb{R}^3$ plane)

In Chapter 12, we discuss functions of several variables (real),

e.g. $z = f(x_1, x_2, \dots, x_n)$. Suppose we denote $\underline{r} = \langle x_1, x_2, \dots, x_n \rangle$,

then $z = f(\underline{r})$ (i.e. scalar valued functions of a vector variable $\underline{r}$)

$\uparrow$ position vector.
i.e. we obtain a scalar field.

Reference: Adam and Essex Chapter 15.1, 15.2, 15.3, 15.4

Larson and Edwards Chapter 15.1-15.2

Vector fields

We will study two types of vector-valued functions: functions that assign a vector to ① a point in the plane, or ② a point in the space. Such functions are called "vector fields". They can be used to represent "force fields" and "velocity fields".

Definition ① A vector field over a plane region R is a function $\underline{F}$ that assigns a vector $\underline{F}(\underline{r}) = \underline{F}(x, y)$ to each point in R .

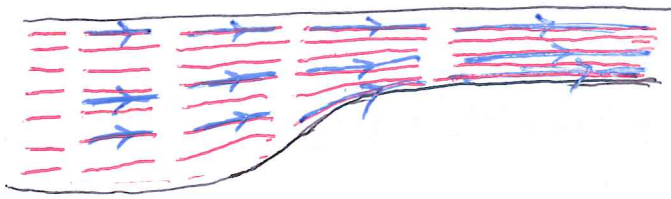
② A vector field over a solid region Q in space is a function $\underline{F}$ that assigns a vector $\underline{F}(\underline{r}) = \underline{F}(x, y, z)$ to each point in Q .

For ①, suppose M and N are two real-valued functions of two variables x and y , defined on a planar region R , the corresponding vector field can be $\underline{F}(x, y) = M(x, y)\underline{\hat{i}} + N(x, y)\underline{\hat{j}}$, where $\underline{r} = \langle x, y \rangle \rightarrow$ position vector

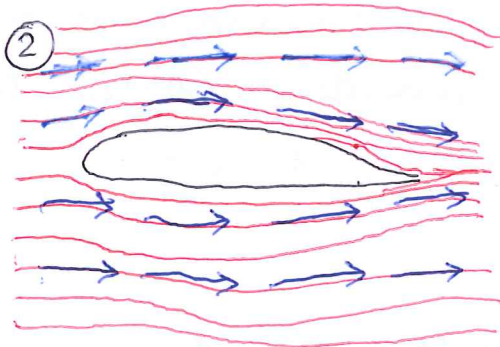
For ②, suppose M , N and P are three real-valued functions of three variables x , y and z , defined in a solid region R in 3-space, the corresponding vector field can be $\underline{F}(x, y, z) = M(x, y, z)\underline{\hat{i}} + N(x, y, z)\underline{\hat{j}} + P(x, y, z)\underline{\hat{k}}$, where $\underline{r} = \langle x, y, z \rangle \rightarrow$ position vector.

NOTE: The vector field is continuous if the component functions M, N and P are P.2 continuous, it is differentiable if each of the component functions is differentiable.

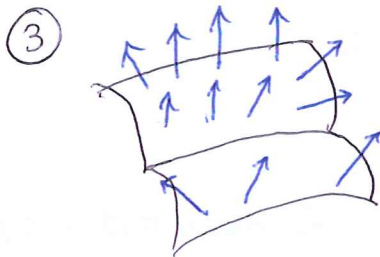
Examples: ① Streamlines in a contracting channel.



Water speeds up as the channel becomes narrower
i.e. velocity vectors increase in length.



Airflow past an inclined airfoil
e.g. velocity vectors of a flow around an airfoil in a wind tunnel.



$F(\underline{r})$
 $F(x, y, z) = c$

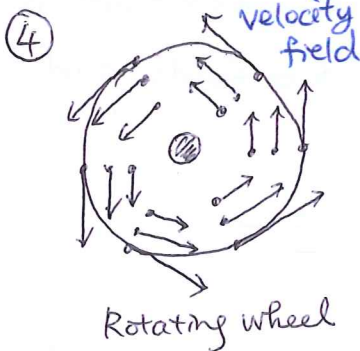
The field of gradient vectors $\underline{\nabla} f$ on a surface $F(x, y, z) = c$.

e.g. If $f(x, y, z) = x^2 + y^2 + z^2$
then the level surface $F(x, y, z) = F(\underline{r})$
is $= f(x, y, z) - x^2 - y^2 - z^2$

$$\underline{\nabla} F = \left\langle \frac{\partial f}{\partial x} - 2x, \frac{\partial f}{\partial y} - 2y, \frac{\partial f}{\partial z} - 2z \right\rangle = \underline{0}$$

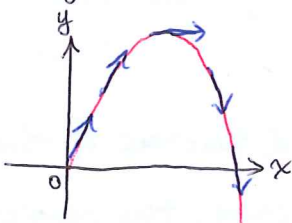
$$= \left(\frac{\partial f}{\partial x} - 2x \right) \underline{i} + \left(\frac{\partial f}{\partial y} - 2y \right) \underline{j} + \left(\frac{\partial f}{\partial z} - 2z \right) \underline{k}$$

is a vector field in space.



Velocity field determined by a wheel rotating on an axle. The farther a point is from the axle, the greater its velocity.

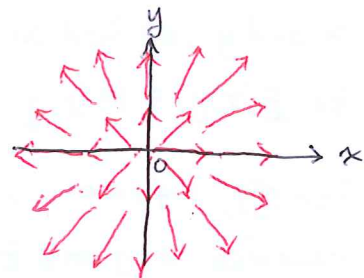
⑤ Projectile motion.



velocity vectors $\underline{v}(t)$ of a projectile's motion make a velocity field along the trajectory.

⑥ Radial field

$\underline{F} = x\underline{i} + y\underline{j}$ of position vectors of points in the plane.



(Remarks: An arrow is drawn with its tail, not the head)

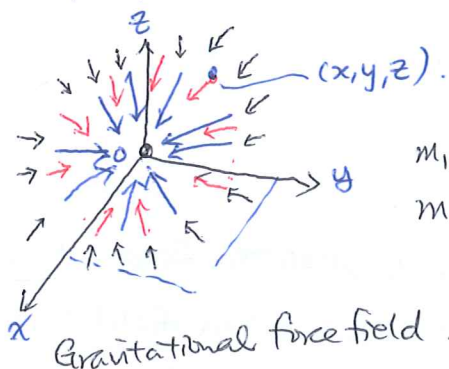
⑦ Gravitational field $\underline{F}(x, y, z)$ due to some object is the force of attraction that the object exerts on a unit mass situated at position (x, y, z) . P.3

e.g. $\underline{F}(x, y, z) = \underline{F}(\underline{r}) = \frac{-Gm_1m_2}{x^2+y^2+z^2} \underline{\hat{u}}$

G : gravitational constant

$\underline{\hat{u}}$: unit vector in the direction from $(0, 0, 0)$ to (x, y, z)

(the force of attraction exerted on a particle of mass m_1 located at (x, y, z) by a particle of mass m_2 located at $(0, 0, 0)$).



m_1 : located at (x, y, z)
 m_2 : located at $(0, 0, 0)$

- $\underline{F}(x, y, z)$ points towards origin
- magnitude of $\underline{F}(x, y, z)$ is the same at all points equidistant from the origin.

Central force field

Suppose the position vector is $\underline{r} = x\underline{\hat{i}} + y\underline{\hat{j}} + z\underline{\hat{k}}$,

then $\underline{F}(\underline{r}) = \frac{-Gm_1m_2}{\|\underline{r}\|^2} \left(\frac{\underline{r}}{\|\underline{r}\|} \right) = \frac{-Gm_1m_2}{\|\underline{r}\|^3} \underline{r}$

Inverse square field

$\underline{F}(\underline{r}) = \frac{k}{\|\underline{r}\|^2} \underline{\hat{u}}$ ($k \in \mathbb{R}$, $\underline{\hat{u}} = \frac{\underline{r}}{\|\underline{r}\|}$: unit vector in the direction of $\underline{r}$)

⑧ Sketch some vectors of the vector field $\underline{F}(x, y) = -y\underline{\hat{i}} + x\underline{\hat{j}}$ in $\mathbb{R}^2$.

Idea: To find level curves in scalar fields, i.e. vectors of equal magnitude lie on circles.

$\|\underline{F}\| = c \Rightarrow x^2 + y^2 = c^2$ (Equation of circle)

e.g. Point

$(1, 0)$

Vector

$\underline{F}(1, 0) = \underline{\hat{j}}$

$(0, 1)$

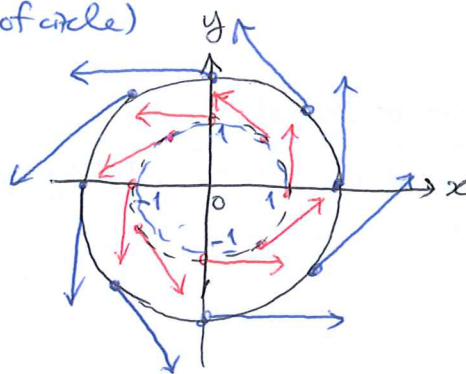
$\underline{F}(0, 1) = -\underline{\hat{i}}$

$(-1, 0)$

$\underline{F}(-1, 0) = -\underline{\hat{j}}$

$(0, -1)$

$\underline{F}(0, -1) = \underline{\hat{i}}$



Each arrow is tangent to a circle with centre at origin

Confirmation: Take the dot product of the position vector $\underline{r} = x\underline{\hat{i}} + y\underline{\hat{j}}$ with the vector $\underline{F}(x, y) = -y\underline{\hat{i}} + x\underline{\hat{j}}$.

$\underline{r} \cdot \underline{F}(x, y) = -xy + xy = 0$

$\therefore \underline{F}(x, y) \perp$ position vector $\underline{r} = \langle x, y \rangle$, i.e. tangent to a circle with centre at $(0, 0)$ and radius $\|\underline{r}\| = \sqrt{x^2 + y^2}$.

$\|\underline{F}(x, y)\| = \sqrt{x^2 + y^2} = \|\underline{r}\|$ (magnitude of vector $\underline{F}(x, y)$ = radius of circle).

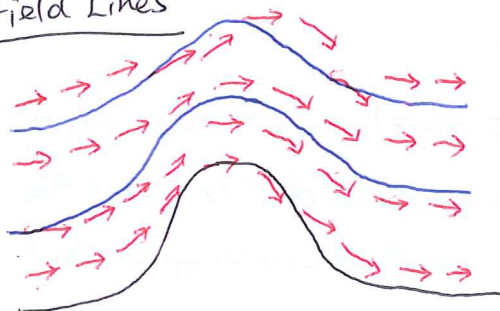
⑨ velocity field of a solid rotating about the z -axis with angular velocity

P.4

$$\underline{\Omega} = \Omega \hat{k} \Rightarrow \underline{v}(x, y, z) = \underline{\Omega} \times \underline{r} = -\Omega y \hat{i} + \Omega x \hat{j}$$

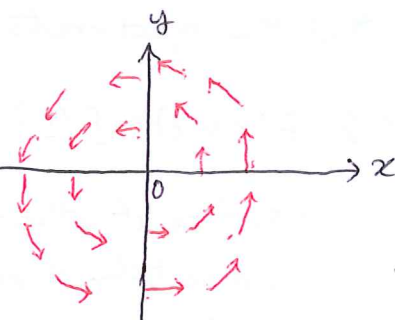
plane vector field

Field Lines



All previous graphical representations of vector fields suggest a pattern of motion through space or in the plane.

Velocity field of a rigid body rotating about the z -axis.



Velocity field and streamlines of wind blowing over a hill

The path will be a curve to which the field is tangent at every point. Such curves are called field lines, or trajectories for the given vector field.

field lines
vector field.

The field lines of $\underline{F}$ only depend on the direction of the field, but NOT depend on the magnitude of $\underline{F}$.

Suppose the field line through some point has parametric equation $\underline{r} = \underline{r}(t)$, then its tangent vector $\frac{d\underline{r}}{dt}$ is parallel to $\underline{F}(\underline{r}(t))$ for all t , i.e. $\frac{d\underline{r}}{dt} = \lambda(t) \underline{F}(\underline{r}(t))$.

Let $\underline{r}(t) = \langle x(t), y(t), z(t) \rangle$, $\underline{F} = \langle F_1, F_2, F_3 \rangle$, then

$$\frac{dx}{F_1(x, y, z)} = \frac{dy}{F_2(x, y, z)} = \frac{dz}{F_3(x, y, z)}$$

Differential equation for the field lines

e.g. Re-visit of Example ⑨

Find the field lines of the velocity field $\underline{v} = \Omega(-y \hat{i} + x \hat{j})$, assume $\Omega, y, x \neq 0$.

Solⁿ $\frac{dx}{-\Omega y} = \frac{dy}{\Omega x} \Leftrightarrow \frac{dx}{-y} = \frac{dy}{x}$

$$\Leftrightarrow x dx = -y dy$$

$$\Leftrightarrow \frac{x^2}{2} = -\frac{y^2}{2} + C$$

$$\Leftrightarrow \boxed{x^2 + y^2 = C_0}$$

field lines are circles centered at the origin in the xy -plane

Regard $\underline{v}$ as a vector field in 3-space, the field lines are horizontal circles centered on the z -axis:

$$\boxed{x^2 + y^2 = C_0, z = C_1}$$

example Find the field lines of the velocity field $\underline{F}(x, y, z) = xz\hat{i} + yz\hat{j} + x\hat{k}$, P.5
 where $x, y, z \neq 0$

Soln: $\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{x}$

Consider $\frac{dx}{x} = \frac{dy}{y} \Leftrightarrow y = C_1 x$ (why?)

and $\frac{dx}{x} = \frac{dz}{z} \Leftrightarrow x = \frac{z^2}{2} + C_2$

Hence the field lines have parametric equations $\begin{cases} x = \frac{t^2}{2} + C_2 \\ y = C_1(\frac{t^2}{2} + C_2) = C_1 \frac{t^2}{2} + C_3 \\ z = t \end{cases}$ $\rightarrow$ parabola $\rightarrow$ parabola
 $t \in \mathbb{R}$.

From previous plots, it seems that the vectors appear to be normal to the level curve from which they emanate. Then, we have a natural question: whether or not the vector field $\underline{F}(x, y)$ or $\underline{F}(x, y, z)$ is the gradient of some differentiable function f ?

Ans: not always possible.

Conservative vector field

Def. A vector field $\underline{F}$ is conservative when there exists a differentiable function f such that $\underline{F} = \nabla f$. We call f as the "potential function" of $\underline{F}$.

Remarks: Potentials are not determined uniquely like anti-derivatives (can have constants).

$\underline{F}$ is conservative in a domain D if and only if $\underline{F} = \nabla f$ at every point of D ; the potential f cannot have any singular points in D .

e.g. ① Every inverse square field is conservative.

let $\underline{F}(x, y, z) = \frac{k}{\|\underline{r}\|^2} \hat{\underline{u}}$ and $f(x, y, z) = \frac{-k}{\sqrt{x^2 + y^2 + z^2}}$, where $\hat{\underline{u}} = \frac{\underline{r}}{\|\underline{r}\|}$

Consider $\nabla f = \frac{kx}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \hat{i} + \frac{ky}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \hat{j} + \frac{kz}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \hat{k}$
 $= \frac{k}{x^2 + y^2 + z^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right) = \frac{k}{\|\underline{r}\|^2} \frac{\underline{r}}{\|\underline{r}\|} = \frac{k}{\|\underline{r}\|^2} \hat{\underline{u}} = \underline{F}$

$\therefore \underline{F}$ is conservative.

② Show that the velocity field $\underline{v} = -\Omega y\hat{i} + \Omega x\hat{j}$ of rigid body rotation about the z -axis is not always conservative.

We try to seek for a potential function $f(x, y)$ for the vector field $\underline{v}(x, y)$.

i.e. $\frac{\partial f}{\partial x} = -\Omega y$ and $\frac{\partial f}{\partial y} = \Omega x \rightarrow f(x, y) = \Omega xy + C_2(x)$

$f(x, y) = -\Omega xy + C_1(y)$

$\therefore -\Omega xy + C_1(y) = \Omega xy + C_2(x)$

$2\Omega xy = C_1(y) - C_2(x)$ for all $(x, y) \in \mathbb{R}^2$.

This is not possible for any choice of the single-variable functions C_1 & C_2 unless $\Omega = 0$.

The following theorem gives a necessary and sufficient condition for a vector field p.6
in the plane to be conservative.

Theorem Let M and N be functions with continuous first partial derivatives on an open disk R . The vector field $\underline{F}(x,y) = M\hat{i} + N\hat{j}$ is conservative if and only if

$$\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

Proof: let us prove the necessary condition at this moment.

Suppose there exists a potential function f such that $\underline{F}(x,y) = \nabla f(x,y)$
 $= M\hat{i} + N\hat{j}$

Then $f_x(x,y) = M$ and $f_y(x,y) = N \Rightarrow f_{xy}(x,y) = \frac{\partial M}{\partial y}$ and $f_{yx}(x,y) = \frac{\partial N}{\partial x}$

Since $f_{xy} = f_{yx}$ (by Clairaut's Theorem), then $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y} \quad \forall (x,y) \in R$.

Example ① Find a potential function for $\underline{F}(x,y) = 2xy\hat{i} + (x^2 - y)\hat{j}$.

Solⁿ From the theorem, $\underline{F}$ is conservative because $\underline{F}_1(x,y) \quad \underline{F}_2(x,y)$

$$\frac{\partial F_1}{\partial y} = 2x \quad \text{and} \quad \frac{\partial F_2}{\partial x} (x^2 - y) = 2x$$

If f is a function whose gradient is equal to $\underline{F}(x,y)$, then

$$\nabla f(x,y) = \underbrace{2xy\hat{i}}_{f_x(x,y)} + \underbrace{(x^2 - y)\hat{j}}_{f_y(x,y)}$$

Consider $f(x,y) = \int f_x(x,y) dx = \int 2xy dx = x^2y + g(y)$.

$$f_y(x,y) = x^2 + g'(y) \Rightarrow \text{i.e. } g'(y) = -y \\ \Rightarrow g(y) = -\frac{y^2}{2} + C.$$

$\therefore f(x,y) = x^2y - \frac{y^2}{2} + C$ gives potential functions of $\underline{F}(x,y)$.

e.g. $f(x,y) = x^2y - \frac{y^2}{2} + 2020$ is a potential function.

Curl and Divergence

① Curl of a vector field $\underline{F}$ is a vector field that gives us at each point, an indication of how the field swirls in the vicinity of that point.

curl of $\underline{F}(x,y,z) = M\hat{i} + N\hat{j} + P\hat{k}$ is $\nabla \times \underline{F}(x,y,z)$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix}$$

$$= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \hat{i} - \left(\frac{\partial P}{\partial x} - \frac{\partial M}{\partial z} \right) \hat{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \hat{k}$$

• If $\text{curl } \underline{F} = \underline{0}$, then $\underline{F}$ is "irrotational"

Theorem (Test for Conservative Vector Field in Space)

P. 7

Suppose that M, N and P are functions with continuous first partial derivatives in an open domain Q in space. The vector field $\underline{F}(x, y, z) = M\hat{i} + N\hat{j} + P\hat{k}$ is conservative if and only if $\nabla \times \underline{F} = \underline{0}$, i.e. $\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}$, $\frac{\partial P}{\partial x} = \frac{\partial M}{\partial z}$ and $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$.

Example Determine whether $\underline{F}(x, y, z) = (xy - \sin z)\hat{i} + (\frac{1}{2}x^2 - \frac{e^y}{z})\hat{j} + (\frac{e^y}{z^2} - x\cos z)\hat{k}$ is conservative in $D = \{(x, y, z) \in \mathbb{R}^3 \mid z \neq 0\}$.

Find a potential if it is.

soln. Note that $\underline{F}$ is not defined when $z=0$. Consider $M(x, y, z) = xy - \sin z$
 $N(x, y, z) = \frac{1}{2}x^2 - \frac{e^y}{z}$
 $P(x, y, z) = \frac{e^y}{z^2} - x\cos z$

$$\frac{\partial P}{\partial y} = \frac{e^y}{z^2} = \frac{\partial N}{\partial z}, \quad \frac{\partial P}{\partial x} = -\cos z = \frac{\partial M}{\partial z} \quad \text{and} \quad \frac{\partial N}{\partial x} = x = \frac{\partial M}{\partial y}.$$

i.e. $\underline{F}$ is conservative in domains that do not intersect the xy -plane.

$$\text{Consider } \begin{cases} \frac{\partial f}{\partial x} = xy - \sin z & \text{--- (1)} \\ \frac{\partial f}{\partial y} = \frac{1}{2}x^2 - \frac{e^y}{z} & \text{--- (2)} \\ \frac{\partial f}{\partial z} = \frac{e^y}{z^2} - x\cos z & \text{--- (3)} \end{cases}$$

from (1), $f(x, y, z) = \frac{1}{2}x^2y - x\sin z + C_1(y, z)$.

Diff. f w.r.t. y , we have

$$f_y(x, y, z) = \frac{1}{2}x^2 + C_{1,y}(y, z)$$

Comparing with (2), $C_{1,y}(y, z) = -\frac{e^y}{z}$

$$C_1(y, z) = -\frac{e^y}{z} + C_2(z).$$

$$\therefore f(x, y, z) = \frac{1}{2}x^2y - x\sin z - \frac{e^y}{z} + C_2(z)$$

Diff. f w.r.t. z , we have

$$f_z(x, y, z) = -x\cos z + \frac{e^y}{z^2} + C_2'(z)$$

By comparing with (3), $C_2'(z) = 0$

$$\Rightarrow C_2(z) = C$$

$$\therefore \text{Potential: } f(x, y, z) = \frac{1}{2}x^2y - x\sin z - \frac{e^y}{z} + C$$

(C may have diff. values in two regions $z > 0$ and $z < 0$ whose union constitutes D).

② Divergence of a vector field is a scalar field that tells us, at each point, the extent to which the field "diverges" or "spreads away" from that point. Let $\underline{F}(x, y) = M\hat{i} + N\hat{j}$

$$\text{div } \underline{F}(x, y) = \nabla \cdot \underline{F}(x, y) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \quad (\text{in plane})$$

$$\text{Let } \underline{F}(x, y, z) = M\hat{i} + N\hat{j} + P\hat{k},$$

$$\text{then } \text{div } \underline{F}(x, y, z) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = \nabla \cdot \underline{F}(x, y, z) \quad (\text{in space}).$$

If $\text{div } \underline{F} = 0$, then $\underline{F}$ is "divergence free".

Example: $\underline{F}(\underline{r}) = y\hat{i} - x\hat{j}$

$$\nabla \cdot \underline{F} = 0 \rightarrow \text{Amount flowing into } \underline{F}$$

$$\nabla \times \underline{F} = -2\hat{k} = \text{amount flowing out of } \underline{F}$$

NOTE: For vector fields that represent velocities of moving particles, divergence p.8 measures the rate of particle flow per unit volume at a point.

In hydrodynamics (study of fluid motion), a divergence free velocity field is called "an incompressible field".

In electricity and magnetism, a divergence free vector field is called "a solenoidal field".

Hint: Let $\underline{F}(x,y,z) = M\hat{i} + N\hat{j} + P\hat{k}$, M, N, P have continuous 2nd partial derivatives

Try to show: ① $\underline{\nabla} \cdot (\underline{\nabla} \times \underline{E}) = 0$ (div of curl of a vector field = 0)

② $\underline{\nabla} \times (\underline{\nabla} f) = \underline{0}$ (curl of gradient of a scalar field = 0)

Def. Laplacian Operator $\underline{\nabla}^2 = \underline{\nabla} \cdot \underline{\nabla} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$
(scalar differential operator) $= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

Line Integrals

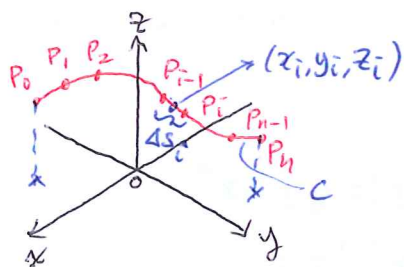
Recall: $\int_a^b f(x) dx$: total amount of a quantity distributed along the x -axis between a and b in terms of line density, $f(x)$ at point x .

$\iint_R f(x,y) dA$ and $\iiint_D f(x,y,z) dV$: total amounts of quantities distributed over regions R in the plane and D in space in terms of areal or volume densities of these quantities.

It may happen that a quantity is distributed with specified line density along a curve in the plane or in 3-space, for which the integration is conducted over a piecewise smooth curve C .

Consider the mass of a wire of finite length, given by a curve C in space.

The density (mass per unit length) of the wire at (x,y,z) is given by $f(x,y,z)$.



$$\text{Total Mass of wire} \approx \sum_{i=1}^n f(x_i, y_i, z_i) \Delta S_i$$

Let $\|\Delta\|$ = length of the longest subarc, and let $\|\Delta\| \rightarrow 0$, limit of this sum approaches the mass of the wire (line integral of f along curve C)

Partition the curve C

by $P_0, P_1, P_2, \dots, P_n$.

Length of the i th subarc = ΔS_i

$$\int_C f(x,y) ds = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta S_i \quad (\text{plane})$$

$$\int_C f(x,y,z) ds = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i, z_i) \Delta S_i \quad (\text{space})$$

provided that limit exists.

To evaluate a line integral over a plane curve C given by $\underline{r}(t) = x(t)\underline{i} + y(t)\underline{j}$, P.9

$$ds = \|\underline{r}'(t)\| dt = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Hence, we have the following theorem

(a) Let $z = f(x, y)$ be continuous in a region containing a smooth curve C . If the plane curve C is traced out by a vector function $\underline{r}(t) = x(t)\underline{i} + y(t)\underline{j}$, where $a \leq t \leq b$,

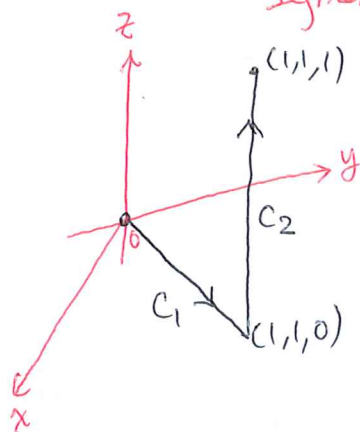
$$\text{then } \int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \left\| \frac{d\underline{r}}{dt} \right\| dt = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

(b) Let $w = f(x, y, z)$ be continuous in a region containing a smooth curve C . If the space curve C is traced out by a vector function $\underline{r}(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$,

$$\text{where } a \leq t \leq b, \text{ then } \int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Examples ① Consider the following path from origin to $(1, 1, 1)$, the union of line segments C_1 and C_2 . Integrate $f(x, y, z) = x - 3y^2 + z$ over $C_1 \cup C_2$.

(from Thomas Calculus)



Solⁿ We choose the simplest parametrizations for C_1 and C_2 .

$$C_1: \underline{r}_1(t) = \langle t, t, 0 \rangle, \text{ where } 0 \leq t \leq 1$$

$$\left\| \frac{d\underline{r}_1}{dt} \right\| = \sqrt{\left(\frac{dt}{dt}\right)^2 + \left(\frac{dt}{dt}\right)^2 + \left(\frac{d0}{dt}\right)^2} = \sqrt{2}$$

$$C_2: \underline{r}_2(t) = \langle 1, 1, t \rangle, \text{ where } 0 \leq t \leq 1$$

$$\left\| \frac{d\underline{r}_2}{dt} \right\| = \sqrt{\left(\frac{d1}{dt}\right)^2 + \left(\frac{d1}{dt}\right)^2 + \left(\frac{dt}{dt}\right)^2} = 1$$

$$\begin{aligned} \text{Then } \int_C f(x, y, z) ds &= \int_{C_1} f(x, y, z) ds + \int_{C_2} f(x, y, z) ds = \int_0^1 f(t, t, 0) \cdot \sqrt{2} dt + \int_0^1 f(1, 1, t) \cdot 1 dt \\ &= \int_0^1 (t - 3t^2) \sqrt{2} dt + \int_0^1 (t - 2) dt \\ &= \sqrt{2} \left[\frac{t^2}{2} - t^3 \right]_0^1 + \left[\frac{t^2}{2} - 2t \right]_0^1 = \frac{-\sqrt{2}}{2} - \frac{3}{2} \end{aligned}$$

② Find the centroid of the circular helix given by $\underline{r}(t) = a \cos t \underline{i} + a \sin t \underline{j} + bt \underline{k}$, where $0 \leq t \leq 2\pi$, a, b are constants.

Solⁿ Consider $ds = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + b^2} dt = \sqrt{a^2 + b^2} dt.$

$$\text{Length of circular helix } L = \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}$$

$$\text{Moment about } z=0 \text{ is } M_{z=0} = \int_C z ds = \int_0^{2\pi} bt \sqrt{a^2 + b^2} dt = 2\pi^2 b \sqrt{a^2 + b^2}$$

$$\Rightarrow z\text{-component of its centroid} = \frac{M_{z=0}}{L} = b\pi.$$

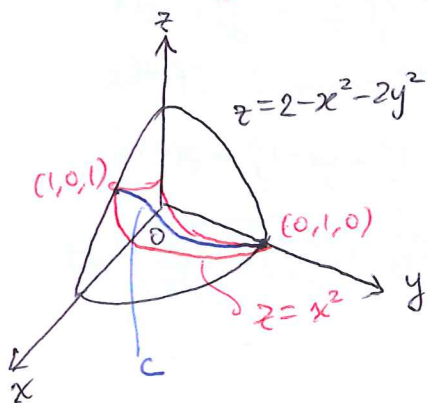
$$\text{Moment about } x=0 \text{ and } y=0 \text{ are } M_{x=0} = \int_C x ds = a \sqrt{a^2 + b^2} \int_0^{2\pi} \cos t dt = 0 \text{ and } M_{y=0} = \int_C y ds = a \sqrt{a^2 + b^2} \int_0^{2\pi} \sin t dt = 0.$$

$$M_y = 0 = \int_C y \, ds = a \sqrt{a^2 + b^2} \int_0^{2\pi} \sin t \, dt = 0$$

P.10

$\therefore$ The centroid is $(0, 0, b\pi)$

- ③ (Adams and Essex §15.3) Find the mass of a wire lying along the first octant part of the curve of intersection of $z = 2 - x^2 - 2y^2$ and $z = x^2$ between $(0, 1, 0)$ and $(1, 0, 1)$, given that the density of the wire at position (x, y, z) is $\delta(x, y, z) = xy$.



Denote C as the first octant part of the curve of intersection.

The curve C lies on the cylinder $z = x^2$, where $x \in [0, 1]$.

$\Rightarrow$ let $x = t, z = t^2$.

$$\therefore 2y^2 = 2 - x^2 - z = 2 - t^2 - t^2 \Rightarrow y^2 = 1 - t^2$$

$y = \sqrt{1 - t^2}$ ($\because C$ lies in the first octant).

$$\therefore x(t) = t, y(t) = \sqrt{1 - t^2}, z(t) = t^2, \text{ where } t \in [0, 1]$$

$$\therefore ds = \sqrt{1 + \frac{t^2}{1-t^2} + 4t^2} \, dt = \frac{\sqrt{1+4t^2-4t^4}}{\sqrt{1-t^2}} \, dt$$

$$\text{Hence, mass of wire} = \int_C xy \, ds = \int_0^1 t \sqrt{1-t^2} \frac{\sqrt{1+4t^2-4t^4}}{\sqrt{1-t^2}} \, dt$$

$$= \int_0^1 t \sqrt{1+4t^2-4t^4} \, dt$$

$$\text{We let } u = t^2, du = 2t \, dt = \int_0^1 \frac{1}{2} \sqrt{1+4u-4u^2} \, du$$

$$= \frac{1}{2} \int_0^1 \sqrt{2-(2u-1)^2} \, du$$

$$\text{let } v = 2u - 1, dv = 2du = \frac{1}{4} \int_{-1}^1 \sqrt{2-v^2} \, dv = \frac{1}{2} \int_0^1 \sqrt{2-v^2} \, dv \text{ (even)}$$

$$= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right) = \frac{\pi+2}{8}$$

(check!)

sub $v = \sqrt{2} \sin w$

Line integrals of vector fields

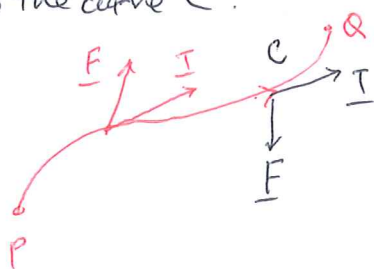
Recall that work done by a constant force of magnitude f in moving an object from a to b along the x -axis $= W = f(b-a)$

If f is variable, then $W = \int_a^b f(x) \, dx$

Now, imagine we have a force field $\underline{F}(x, y, z) = \underline{F}(\underline{r}) = M(x, y, z)\hat{i} + N(x, y, z)\hat{j} + P(x, y, z)\hat{k}$, the work done by this force field in moving an object along a smooth curve C given by $\underline{r}(t) = \langle x(t), y(t), z(t) \rangle$, where $t \in [a, b]$ is defined as follows:

(in space)

$\int_C \underline{F}(x,y,z) \cdot \underline{T}(x,y,z) ds$, where $\underline{T}(x,y,z) = \frac{\underline{r}'(t)}{\|\underline{r}'(t)\|}$ is the unit tangent vector to the curve C .



$$W = \int_C \underline{F} \cdot d\underline{r} = \int_a^b \underline{F} \cdot \underline{r}'(t) dt \quad (\text{scalar}).$$

This line integral changes sign if the orientation of C is reversed, and is independent of parametrization of C .

We have 6 possible ways to represent the work done integrals.

① $\int_C \underline{F}(x,y,z) \cdot \underline{T}(x,y,z) ds$ ② $\int_a^b \underline{F}(x(t), y(t), z(t)) \cdot \frac{\underline{r}'(t)}{\|\underline{r}'(t)\|} \|\underline{r}'(t)\| dt$

③ $\int_a^b \underline{F}(x(t), y(t), z(t)) \cdot \underline{r}'(t) dt$ ④ $\int_C \underline{F}(x,y,z) \cdot d\underline{r}$

⑤ $\int_a^b \left[M(x(t), y(t), z(t)) \frac{dx}{dt} + N(x(t), y(t), z(t)) \frac{dy}{dt} + P(x(t), y(t), z(t)) \frac{dz}{dt} \right] dt$

⑥ $\int_C M dx + N dy + P dz$

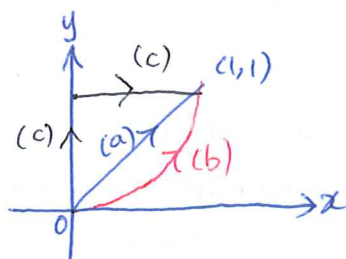
Remarks: If C is a closed curve, $\oint_C \underline{F} \cdot d\underline{r}$ denotes the circulation of $\underline{F}$ around the closed curve C .

Examples ① Let $\underline{F}(x,y) = y^2 \underline{i} + 2xy \underline{j}$.

Evaluate $\int_C \underline{F} \cdot d\underline{r}$ from $(0,0)$ to $(1,1)$ along (a) the straight line $y=x$

(b) the curve $y=x^2$

(c) the piecewise smooth path that consists of the line segments from $(0,0)$ to $(0,1)$, then from $(0,1)$ to $(1,1)$.



(a) Parametrize the straight path (a) as $\underline{r}(t) = t\underline{i} + t\underline{j}$ ($0 \leq t \leq 1$)

$$\therefore \underline{F} \cdot d\underline{r} = (t^2 \underline{i} + 2t^2 \underline{j}) \cdot (\underline{i} + \underline{j}) dt = 3t^2 dt$$

$$\int_C \underline{F} \cdot d\underline{r} = \int_0^1 3t^2 dt = t^3 \Big|_0^1 = \underline{\underline{1}}$$

(b) Parametrize the parabola as $\underline{r}(t) = t\underline{i} + t^2 \underline{j}$, where $0 \leq t \leq 1$.

$$\therefore \underline{F} \cdot d\underline{r} = (t^4 \underline{i} + 2t^3 \underline{j}) \cdot (\underline{i} + 2t \underline{j}) dt = 5t^4 dt$$

$$\int_C \underline{F} \cdot d\underline{r} = \int_0^1 5t^4 dt = t^5 \Big|_0^1 = \underline{\underline{1}}$$

(c) For the vertical segment, use y as the parameter ($x=0, dx=0$)

For the horizontal segment, use x as the parameter ($y=1, dy=0$)

$$\int_C \underline{F} \cdot d\underline{r} = \int_C y^2 dx + 2xy dy = \int_0^1 0 dy + \int_0^1 1 dx = \underline{\underline{1}}$$

same
value
along all
3 paths

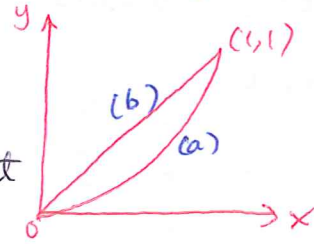
($\because \underline{F}(x,y)$
is conservative)
on a simply
connected
domain

② Find $\int_C \underline{F} \cdot d\underline{r}$, where $\underline{F}(x,y) = e^{x-1} \hat{i} + xy \hat{j}$, and C is given by

(a) $\underline{r}(t) = t^2 \hat{i} + t^3 \hat{j}$, where $0 \leq t \leq 1$

(b) $\underline{r}(t) = t \hat{i} + t \hat{j}$, where $0 \leq t \leq 1$

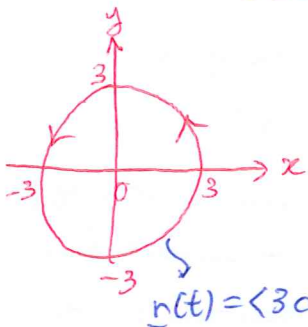
Solⁿ (a) $\int_C \underline{F} \cdot d\underline{r} = \int_0^1 (e^{t^2-1} \hat{i} + t^5 \hat{j}) \cdot (2t \hat{i} + 3t^2 \hat{j}) dt$
 $= \int_0^1 (2te^{t^2-1} + 3t^7) dt$
 $= \left[e^{t^2-1} + \frac{3}{8} t^8 \right]_0^1 = \frac{11}{8} - \frac{1}{e}$



(b) $\int_C \underline{F} \cdot d\underline{r} = \int_0^1 (e^{t-1} \hat{i} + t^2 \hat{j}) \cdot (\hat{i} + \hat{j}) dt = \int_0^1 (e^{t-1} + t^2) dt$
 $= \left[e^{t-1} + \frac{1}{3} t^3 \right]_0^1 = \frac{4}{3} - \frac{1}{e}$

The line integral depends on the path from $(0,0)$ to $(1,1)$ along which the integral is evaluated. $\therefore$ The vector field is NOT conservative.

③ Let C be the circle of radius 3 given by $\underline{r}(t) = 3 \cos t \hat{i} + 3 \sin t \hat{j}$, where $0 \leq t \leq 2\pi$. Evaluate $\int_C y^3 dx + (x^3 + 3xy^2) dy$.



$x(t) = 3 \cos t, y(t) = 3 \sin t$

$dx = -3 \sin t dt, dy = 3 \cos t dt$

We use the formula

$\int_C M dx + N dy$, where $M(x,y) = y^3$ and $N(x,y) = x^3 + 3xy^2$
 $= \int_C (27 \sin^3 t)(-3 \sin t) dt + (27 \cos^3 t + 8(3 \cos t \sin^2 t)(3 \cos t) dt$
 $= 81 \int_0^{2\pi} (\cos^4 t - \sin^4 t + 3 \sin^2 t \cos^2 t) dt$
 $= 81 \int_0^{2\pi} (\cos^2 t - \sin^2 t + \frac{3}{4} \sin^2 2t) dt$
 $= 81 \int_0^{2\pi} \left[\cos 2t + \frac{3}{4} \left(\frac{1 - \cos 4t}{2} \right) \right] dt$
 $= 81 \left[\frac{\sin 2t}{2} + \frac{3}{8} t - \frac{3}{32} \sin 4t \right]_0^{2\pi} = \underline{\underline{\frac{243}{4} \pi}}$

Remarks: The orientation of C affects the value of the line integral.

Denote $-C$ as the curve with opposite orientation as C , then

$\int_{-C} M dx + N dy = - \int_C M dx + N dy.$